

5-1 · Determine the Area of Parallelograms and Rhombuses

Connect & apply · 8 tasks · Prepares for 6.G.A.1

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

1. FOUNDATION 1

Compute 24×13 .

2. FOUNDATION 2

A rectangle is 8 cm long and 6 cm wide. Find its area in square centimeters.

3. FOUNDATION 3

A parallelogram has base 10 cm, slanted side 7 cm, and perpendicular height 4 cm. Which number is the height?

4. FOUNDATION 4

A rectangle measures 3 meters by 5 meters. State its area with units.

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

5-1 · Determine the Area of Parallelograms and Rhombuses

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Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

5. CHECK A

Area of an 8 cm by 5 cm rectangle? In a parallelogram, is the height the slanted side?

6. CHECK B

Area of a 9 cm by 7 cm rectangle? In a parallelogram, is the height the slanted side?

7. CHECK C

Area of a 12 m by 4 m rectangle? A parallelogram has base 12 m, slant side 5 m, height 4 m. What is its area?

8. TRANSFER

Two parallelograms both have base 10 cm and height 6 cm but different slant sides. Compare their areas.

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

5-2 · Determine the Area of Triangles

Connect & apply · 8 tasks · Prepares for 6.G.A.1

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

1. FOUNDATION 1

Compute $(14 \times 9) \div 2$.

2. FOUNDATION 2

Compute $\frac{1}{2} \times 38$.

3. FOUNDATION 3

A parallelogram has base 12 cm and height 5 cm.
Find its area in square centimeters.

4. FOUNDATION 4

A triangle has base 8 cm, perpendicular height 6 cm, and slanted side 10 cm. Which length belongs in the height part of its area formula?

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

5-2 · Determine the Area of Triangles

Connect & apply · 8 tasks · Prepares for 6.G.A.1

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

5. CHECK A

7×4 , then half? Area of a triangle with base 7 m and height 4 m?

6. CHECK B

9×6 , then half? Area of a triangle with base 9 cm and height 6 cm?

7. CHECK C

12×5 , then half? Area of a triangle with base 12 in and height 5 in?

8. TRANSFER

A triangle has area 36 cm^2 and base 12 cm. What is its height?

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

5-3 · Determine the Area of Trapezoids

Connect & apply · 8 tasks · Prepares for 6.G.A.1

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

1. FOUNDATION 1

Compute $(7 + 11) \times 4$.

2. FOUNDATION 2

Compute $\frac{1}{2} \times 54$.

3. FOUNDATION 3

Evaluate $\frac{1}{2} \times (6 + 10) \times 5$.

4. FOUNDATION 4

A shape is made of a rectangle of area 24 and two triangles of area 6 each, without overlap. Find the total area.

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

5-3 · Determine the Area of Trapezoids

Connect & apply · 8 tasks · Prepares for 6.G.A.1

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

5. CHECK A

$\frac{1}{2} \times (6 + 10) \times 4 = ?$ Do the parentheses first.

6. CHECK B

$\frac{1}{2} \times (8 + 12) \times 5 = ?$ Do the parentheses first.

7. CHECK C

$\frac{1}{2} \times (7 + 9) \times 6 = ?$ Do the parentheses first.

8. TRANSFER

A trapezoid has bases 6 and 10 and area 48.
What is its height?

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

5-4 · Apply Area Concepts to Solve Problems

Connect & apply · 8 tasks · Prepares for 6.G.A.1

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

1. FOUNDATION 1

Find the area of a triangle with base 10 cm and height 8 cm, in square centimeters.

2. FOUNDATION 2

A rectangle has area 72 square meters. A 15-square-meter section is removed. What area remains?

3. FOUNDATION 3

A total side length is 14 cm. One part is 9 cm. How long is the other part?

4. FOUNDATION 4

Six identical tiles each cover 12 square centimeters. What is the total area?

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

5-4 · Apply Area Concepts to Solve Problems

Connect & apply · 8 tasks · Prepares for 6.G.A.1

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

5. CHECK A

A 10 by 6 rectangle has a 3 by 2 rectangle cut out. What area is left?

6. CHECK B

A 12 by 8 rectangle has a 4 by 3 rectangle cut out. What area is left?

7. CHECK C

A 9 by 9 square has a triangle of base 6 and height 4 cut out. What area is left?

8. TRANSFER

A rectangular garden 15 m by 10 m has a 3 m by 3 m square shed inside. How much garden is left to plant?

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

5-5 • Determine the Volume of Rectangular Prisms

Connect & apply · 8 tasks · Prepares for 6.G.A.2

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

1. FOUNDATION 1

Compute $4 \times 7 \times 5$.

2. FOUNDATION 2

Compute $1\frac{1}{2} \times \frac{2}{3}$.

3. FOUNDATION 3

A rectangular base is 9 cm by 4 cm. Find its area in square centimeters.

4. FOUNDATION 4

A prism measures 2 cm by 3 cm by 5 cm. State its volume with units.

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

Connect & apply · 8 tasks · Prepares for 6.G.A.2

Name: _____ Date: _____

5. CHECK A

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

5-6 · Represent Three-Dimensional Figures in Two Dimensions

Connect & apply · 8 tasks · Prepares for 6.G.A.4

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

1. FOUNDATION 1

How many faces, edges, and vertices does a rectangular prism have?

2. FOUNDATION 2

How many faces does a square pyramid have, including its base?

3. FOUNDATION 3

A net has a rectangular face 3 cm by 7 cm. Find that face's area in square centimeters.

4. FOUNDATION 4

A prism has face pairs sized 3 by 4, 3 by 5, and 4 by 5. How many faces are 3 by 4?

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

5-6 · Represent Three-Dimensional Figures in Two Dimensions

Connect & apply · 8 tasks · Prepares for 6.G.A.4

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

5. CHECK A

How many faces does a rectangular prism have?
How many matching pairs is that?

6. CHECK B

How many faces does a triangular prism have?
How many are triangles?

7. CHECK C

How many faces does a square pyramid have?
How many are triangles?

8. TRANSFER

A net has 2 triangles with base 6 and height 4,
and 3 rectangles each 6 by 10. What solid is it,
and what is its surface area?

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

5-7 · Determine Surface Area of Prisms

Connect & apply · 8 tasks · Prepares for 6.G.A.4

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

1. FOUNDATION 1

Add the face areas $12 + 12 + 15 + 15 + 20 + 20$.

2. FOUNDATION 2

Find the area of a triangle with base 6 cm and height 9 cm, in square centimeters.

3. FOUNDATION 3

In a rectangular prism with dimensions 2, 3, 5, list the three different face areas.

4. FOUNDATION 4

One face has area 18 square centimeters. What is the combined area of it and its matching opposite face?

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

5-7 · Determine Surface Area of Prisms

Connect & apply · 8 tasks · Prepares for 6.G.A.4

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

5. CHECK A

Faces measure 12, 12, 20, 20, 15, 15 square units — what is the surface area?

6. CHECK B

Faces measure 18, 18, 24, 24, 30, 30 square units — what is the surface area?

7. CHECK C

A prism has faces 10, 10, 14, 14, 35, 35 square units — what is the surface area?

8. TRANSFER

A cube has surface area 150 cm². What is the length of one edge?

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

5-8 · Determine Surface Area of Pyramids

Connect & apply · 8 tasks · Prepares for 6.G.A.4

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

1. FOUNDATION 1

Find the area of a triangle with base 8 cm and perpendicular height 7 cm, in square centimeters.

2. FOUNDATION 2

Find the area of a square base with side 6 cm, in square centimeters.

3. FOUNDATION 3

Four congruent triangular faces each have area 21 square centimeters. Find their combined area.

4. FOUNDATION 4

A pyramid's vertical height is 9 cm. The altitude drawn within a triangular face is 10 cm. Which height finds that face's area?

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help

5-8 · Determine Surface Area of Pyramids

Connect & apply · 8 tasks · Prepares for 6.G.A.4

Name: _____ Date: _____

Show your thinking. Label units. Use a sketch, words, or equations. Take the time you need.

5. CHECK A

One triangular face is 24 square inches. With four identical faces, what is the lateral area?

6. CHECK B

One triangular face is 30 square inches. With four identical faces, what is the lateral area?

7. CHECK C

One triangular face is 18 cm² and the square base is 36 cm². What is the total surface area of the 4-face pyramid?

8. TRANSFER

A square pyramid has base edge 10 cm and slant height 12 cm. Find the total surface area.

One strategy I used: _____

My next step: ☐ Explain it to someone ☐ Try another example ☐ Ask for help